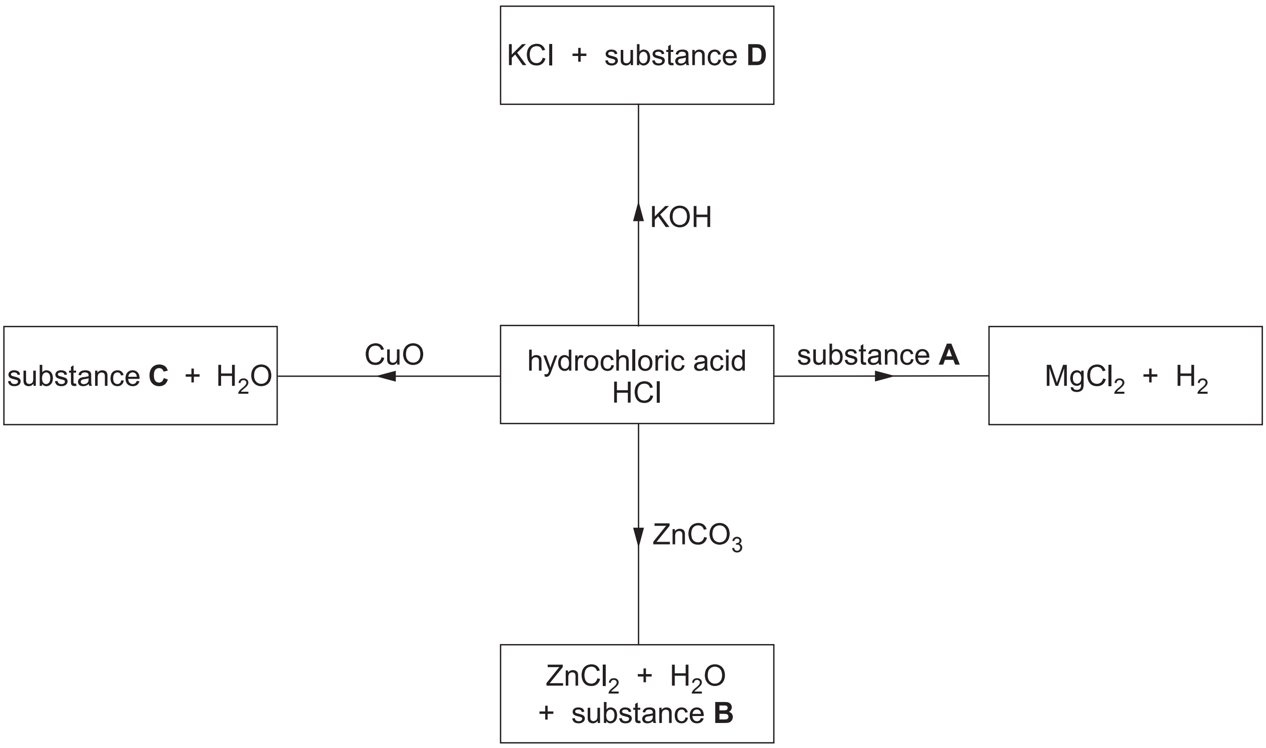
2. The reactions of acids with metals, bases and carbonates are summarised in the following equations.

- acid + metal → salt + hydrogen
- acid + base → salt + water
- acid + carbonate → salt + water + carbon dioxide

(a) The diagram shows some reactions of hydrochloric acid, HCl.



(i) Give the **names** of substances **A** and **B**. [2]

Substance **A** ....

Substance **B** ....

(ii) Give the **formulae** of substances **C** and **D**. [2]

Substance **C** ....

Substance **D** ....

Examiner only

(b) Complete the equation for the reaction between hydrochloric acid and sodium carbonate by

- writing the formula of sodium carbonate on the dotted line
- putting a number into the box to balance the equation [2]

$\square$ HCl + ... $\longrightarrow$ 2NaCl + H_2O + CO_2

(c) Silver nitrate solution is used to identify the chloride ions present in hydrochloric acid.

(i) Give the observation made when silver nitrate solution is added to hydrochloric acid. [1]

....

(ii) Put a tick (✓) in the box next to the correct ionic equation for the reaction between silver nitrate and hydrochloric acid. [1]

$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(aq)$	$\square$
$Ag^-(aq) + Cl^+(aq) \longrightarrow AgCl(aq)$	$\square$
$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$	$\square$
$Ag^+(s) + Cl^-(s) \longrightarrow AgCl(s)$	$\square$
$Ag^-(aq) + Cl^+(aq) \longrightarrow AgCl(s)$	$\square$

3430UE01
05

8